Yue Cao

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Employment

Syracuse University, Syracuse, NY *2024 Aug - Present*
Assistant Teaching Professor in the Department of Electrical Engineering and Computer Science

Education

Purdue University, West Lafayette, IN *2016 - 2024*
PhD in Electrical and Computer Engineering
Advisor: Prof. C. S. George Lee
Thesis: "Towards Manipulator Task-Oriented Programming: Automating Behavior-Tree Configuration,"

Ohio State University, Columbus, OH *2014 - 2016*
M.S. in Electrical and Computer Engineering

Shanghai Jiao Tong University, Shanghai, China *2011 - 2015*
B.S. in Automation Engineering

Publication

Yue Cao, "GitHub Repository as Handbook, Jupyter Notebook as Report: A Format Redesign for the Electrical Engineering II Lab," in *Proceedings of the Annual Conference and Exposition* (*Accepted*), Charlotte, NC, 2026

Yue Cao and Jiamin Zhao, "Embedding Intelligence: Teaching Modern AI Implementation in a Raspberry Pi–Centric Laboratory Course," in *Proceedings of the ASEE St. Lawrence Section Conference* (*Accepted*), Ithaca, NY, 2026

Yue Cao, "Large Language Models for Nodal Analysis in Circuits Education: An Evaluation," in *Proceedings of the IEEE SoutheastCon*, Huntsville, AL, 2026

Yue Cao, "Performance Evaluation of Whisper-Series Speech Transcription Models on Raspberry Pi," in *Proceedings of the ACM/IEEE Symposium on Edge Computing – Edge Intelligence Workshop*, Arlington, VA, 2025

Yue Cao, "Performance Evaluation of Whisper-Series Speech Transcription Models on Raspberry Pi," in *Proceedings of the ACM/IEEE Symposium on Edge Computing – Edge Intelligence Workshop*, Arlington, VA, 2023

Yue Cao and C. S. George Lee, "Ground Manipulator Primitive Tasks to Executable Actions using Large Language Models," in *Proceedings of the AAAI Fall Symposium on Unifying Representations for Robot Application Development*, Arlington, VA, 2023

Yue Cao and C. S. George Lee, "Robot Behavior-Tree-Based Task Generation with Large Language Models," in *Proceedings of the AAAI Spring Symposium on Challenges Requiring the Combination of Machine Learning and Knowledge Engineering (AAAI-MAKE)*, San Francisco, CA, 2023.

Yue Cao and C. S. George Lee, "Behavior-Tree Embeddings for Robot Task-Level Knowledge," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Kyoto, Japan, 2022.

Haoan Wang, **Yue Cao**, Bilin Aksun Güvenç, Levent Güvenç, "MPC Based Automated Steering of a Low Speed Shuttle for Socially Acceptable Accident Avoidance," in *Proceedings of the ASME Dynamic Systems and Control Conference (DSCC)*, Minneapolis, MN, 2016.

Teaching

Assistant Teaching Professor, Syracuse University, NY *2024 - Present*

Courses:
SCN 147 – **EE I: Circuits and Robots (Pre-College)** *26*
ELE 231 – **Electrical Engineering Fundamentals** *25F, 26F*
ELE 292 – **Linear Systems Laboratory** *24F, 25S, 25F, 26F*
ELE 251 – **Fundamentals of Linear Systems** *25S*
CSE 398 – **Embedded and Mobile Systems Laboratory** *25S, 26S*
CSE 400 – **Intelligent Robotics** *26S*

Course Material Samples are available on:
`https://botyue.github.io/teaching.html`

Instructor, Purdue University, West Lafayette, IN *2016 - 2024*
ECE 308 – **Systems Simulation and Control Lab**
Collaborate with **Prof. Shreyas Sundaram** to design curriculum.

Purdue Magoon Excellence in Teaching Award *2019*
Selected by faculty and students to recognize graduate students who were exemplary in teaching.

Purdue Teaching Academy Graduate Teaching Award *2023*
Selected by department from every 50 graduate teaching assistants.

Presentation

Poster, "Towards Manipulator Task-Oriented Programming: Automating Behavior-Tree Configuration", at *the Northeast Robotics Colloquium (NERC)*, Ithaca, NY, 2025.

Oral Presentation, "Ground Manipulator Primitive Tasks to Executable Actions using Large Language Models", as Late-Breaking Results at *the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Detroit, MI, 2023.

Oral Presentation, "Robot Behavior-Tree-Based Task Generation with Large Language Models", at *the Purdue Robotics and Intelligent Systems Expo*, West Lafayette, IN, 2023.

Service/Outreach

Department Committee for Computer Science program ABET *2025-2026*

Booth Host, Mircon Day at Syracuse University *2026*

Event Coordinator, Syracuse School District Vex Robotics Tournament *2026*

Booth Host, ECS College open house, ECS College showcase *2025-2026*

Judge, *Purdue Undergraduate Research Conference* *2018, 2023*

Demo, K-12 student outreach for Lego Robotics classes *2017*rant *2017, 2018, 2023*

Other Awards

"Shared Competencies" Assignment Design Award, Syracuse University Center for Teaching and Learning Excellence *2026*

Summer Institute for Technology-Enhanced Teaching and Learning, Syracuse University ITS *2026*

Course Redesign Institute, Syracuse University Center for Teaching and Learning Excellence *2025*

Purdue College of Engineering graduate conference travel grant *2023*

Purdue Graduate School summer research grant *2017, 2018, 2023*

Review

Reviewer, *IEEE Robotics and Automation Letters (RA-L)* *2026*

Reviewer, *IEEE Int. Conf. Tools with Artificial Intelligence (ICTAI)* *2026*

Reviewer, *IEEE Int. Conf. Systems, Man, and Cybernetics (SMC)* *2026*

Reviewer, *IROS 2026* *2026*

Reviewer, *IROS 2025 Workshop on Embodied AI and Robotics for Future Scientific Discovery* *2025*

Reviewer, *RSS 2024 Workshop on Task Specification for General-Purpose Intelligent Robots* *2024*

Reviewer, *ACL Rolling Review* *2024*

Reviewer, in the Young Reviewers Program of *the IEEE Robotics and Automation Society (RAS)* *2023 - present*

Reviewer, *IEEE International Conference on Robotics and Automation (ICRA)* *2023*

Reviewer, *IEEE Robotics and Automation Letters (RA-L)* *2023*

Reviewer, *IEEE-RAS International Conference on Humanoid Robots* *2023*

Professional Experience

Graduate Researcher, Purdue University, West Lafayette, IN *2016 - 2024*
with **Prof. C. S. George Lee**, at the Assistive Robotics Technology Laboratory
Focusing on utilizing the AI techniques to facilitate the intelligence of robot systems.
Researching on improving behavior-tree-based planning, including efficient task re-usage, autonomous generation of hierarchical task.
Researching on robot planning-to-acting transition, including action grounding of manipulator primitive tasks and reinforcement learning within precedence-constrained tasks.

Research Intern, Bosch Research and Technology Center, Shanghai, China *2016*
Surveyed on the real-time 3-D human motion tracking using inertial sensors.

Graduate Researcher, Ohio State University, Columbus, OH *2015 - 2016*
with **Prof. Levent Güvenç**, at the Automated Driving Lab.
Developed an approach to generate vehicle simulation trajectories based on OpenStreetMap.
Investigated the realization of autonomous driving on Ford Fusion hybrid vehicles.

Graduate Grader, Ohio State University, Columbus, OH *2015*
with **Prof. Yuan F. Zheng**
Course: ECE 5463 – Introduction to Real-Time Robotics Systems.

Software Engineer Intern, Honeywell Security Group, Shanghai, China *2015*
Function test for the Honeywell video storage team.

Undergraduate Researcher, Shanghai Jiao Tong University, China
with **Prof. Ning Li**, at the Key Lab of System Control and Information Processing
Model predictive controller design for HVAC systems.